

```

create table employee(
    e_id int primary key,
    e_name varchar(50),
    email varchar(100),
    salary decimal(10,2),
    d_id int,
    foreign key (d_id) references department(d_id)
);

insert into department values
(1,"CSE"),
(2,"CSE-AI"),
(3,"CSE-DS"),
(4,"CSE-CS"),
(5,"BCA");

insert into employee values
(1,"A","A@gmail.com",10000.25,1),
(2,"B","B@gmail.com",20000.05,4),
(3,"C","C@gmail.com",25000.75,2),
(4,"D","D@gmail.com",25000.45,3),
(5,"E","E@gmail.com",16000.23,1),
(6,"F","F@gmail.com",37000.25,1),
(7,"G","G@gmail.com",26000.05,4),
(8,"H","H@gmail.com",34000.75,3),
(9,"I","I@gmail.com",23000.45,2),
(10,"J","J@gmail.com",17000.23,2);

-- select * from department;
-- select * from employee;

-- 1. find employees who earn more than the average salary of their department
select * from employee e where salary > (
    select AVG(salary) from employee where d_id=e.d_id
);

-- 2. list employees who are the highest paid in their respective departments
select * from employee e where salary = (
    select MAX(salary) from employee where d_id=e.d_id
);

-- 3. display departments with more than 2 employees
select d_id,count(*) from employee group by d_id having count(*) > 2;

-- 4. list employees whose salary is above the overall average but not the highest in their department
select * from employee e where salary > (
    select avg(salary) from employee
)
and salary < (
    select MAX(salary) from employee where d_id=e.d_id
);

-- 5. find employees who do not belong to departments having any employee with salary less than 20000
select * from employee e where e.d_id not in (
    select d_id from employee where salary < 20000
);

```

4 08:38:06 create table department(d_id int primary key, d_name varchar(10))
 5 08:38:06 create table employee(e_id int primary key, e_name varchar(50), email varchar(100), salary decimal(10,2), d_id int, foreign key (d_id) references department(d_id))
 6 08:38:06 insert into department values (1,"CSE"), (2,"CSE-AI"), (3,"CSE-DS"), (4,"CSE-CS"), (5,"BCA")
 7 08:38:06 insert into employee values (1,"A","A@gmail.com",10000.25,1), (2,"B","B@gmail.com",20000.05,4), (3,"C","C@gmail.com",25000.75,2), (4,"D","D@gmail.com",25000.45,3)
 8 08:38:06 select * from employee e where salary > (select AVG(salary) from employee where d_id=e.d_id) LIMIT 0, 1000
 9 08:38:06 select * from employee e where salary = (select MAX(salary) from employee where d_id=e.d_id) LIMIT 0, 1000
 10 08:38:06 select d_id,count(*) from employee group by d_id having count(*) > 2 LIMIT 0, 1000
 11 08:38:06 select * from employee e where salary > (select avg(salary) from employee) and salary < (select MAX(salary) from employee where d_id=e.d_id) LIMIT 0, 1000
 12 08:38:06 select * from employee e where e.d_id not in (select d_id from employee where salary < 20000) LIMIT 0, 1000

Result Grid | Filter Rows: | Edit:

	e_id	e_name	email	salary	d_id
▶	3	C	C@gmail.com	25000.75	2
	6	F	F@gmail.com	37000.25	1
	7	G	G@gmail.com	26000.05	4
	8	H	H@gmail.com	34000.75	3
	9	I	I@gmail.com	23000.45	2
*	NULL	NULL	NULL	NULL	NULL

employee 1 × employee 2 Result 3 employee 4 employee 5

Result Grid | Filter Rows: | Edit: | Export:

	e_id	e_name	email	salary	d_id
▶	3	C	C@gmail.com	25000.75	2
	6	F	F@gmail.com	37000.25	1
	7	G	G@gmail.com	26000.05	4
	8	H	H@gmail.com	34000.75	3
*	NULL	NULL	NULL	NULL	NULL

employee 1 employee 2 × Result 3 employee 4 employee 5

Result Grid | Filter Rows: | Edit:

	d_id	count(*)
▶	1	3
	2	3

Result 3 × employee 4

Result Grid | Filter Rows: | Edit:

	e_id	e_name	email	salary	d_id
▶	4	D	D@gmail.com	25000.45	3
*	NULL	NULL	NULL	NULL	NULL

employee 4 × employee 5

Result Grid | Filter Rows: | Edit:

	e_id	e_name	email	salary	d_id
▶	2	B	B@gmail.com	20000.05	4
	4	D	D@gmail.com	25000.45	3
	7	G	G@gmail.com	26000.05	4
	8	H	H@gmail.com	34000.75	3
*	NULL	NULL	NULL	NULL	NULL

employee 4 employee 5 ×